ALLIED WELDING SYSTEMS

What It Costs, and Why

Where the $6,000 comes from, what you pay to keep it running, and what it would cost if we moved the question-answering assistant onto a paid online service.

The short answers

10 wks — **To pay for itself**, at ten orders a week. Two hours of hand-work per order becomes about five minutes. Everything below is the arithmetic behind that sentence.

$6,000 — **To install it.** About 63 hours of work at $95 an hour. Published rates for this kind of work run $100 to $175.

$15–100 — **A year to run it.** Electricity and a backup drive. There is no software fee and no charge per order.

$840 — **A year you can save.** The system works on the cheaper Monday plan. Nobody needs to buy the expensive one.

$3–68 — **A year, if the AI assistant moved online.** Cheaper than the computer you would otherwise buy to run it here. The reason to say no is not the money.

> **Nothing here is a round number someone picked**
>
> The work hours come from counting what is actually being installed. The running figures were measured on a real machine. The online-assistant figures come from capturing the actual questions the system sends and pricing them. Vendor prices were read off the suppliers' own pages on 24 October 2026.

1. Why installing it costs $6,000

What is actually being installed

The fee is not for writing the software — that is done. It is for putting something this size into live use at Allied. The size sets the work.

Part	Files	Lines of instruction	What it is
The engine	67	13,622	Orders, invoice checking, compliance, Sage and Monday links
The checks	27	5,000	150 automatic tests that run every time anything changes
The screens	37	11,439	Order desk, review list, compliance records, assistant
Total	**131**	**30,061**	

Sixteen areas of function, five record types, two outside systems, four levels of staff access.

> **Why that matters:** Installing a system of this size is not a morning's work because the risk is not in the installing — it is in your real parts list, your real customers and your real Sage numbering behaving differently from test data. Most of the 63 hours is finding and settling those differences before they reach a live order.

Why $95 an hour

Two comparisons, because the rate sits deliberately between them.

Published 2026 rates	Range
American IT consultants (Cleveroad)	$100–175 an hour
American IT consultants (BTI)	$100–250 an hour
Accounting-system specialists (ERP Focus)	$150–175 an hour
Independent freelancers, US average (ZipRecruiter)	$47.71 an hour
This job	**$95 an hour**

What you are not getting at either end of that table

The two options above fail Allied in opposite ways, and the difference is almost entirely about what happens **after** the install, which is when a system like this either becomes part of how you work or quietly stops being used.

	A consultancy	A cheap freelancer	This arrangement
Knows your people and how Allied actually runs	No	No	Yes
On site, in the building	Billed as travel	Remote	Yes
Who you reach when it breaks	A ticket queue	Whoever they are working for now	Me, directly
Still around in two years	The firm, not the person	Unlikely	Yes
Has to live with the result	No	No	Yes

The part that does not appear on any rate card

I know who is in this business and what each person actually does with their day. That is why the review screens are laid out the way they are, why the system refuses the things it refuses, and why the training is two short sessions rather than a manual. **A stranger at $150 an hour would have spent a chunk of your money learning what I already know**, and would still have guessed wrong about the parts of the job nobody writes down.

> **Why that matters:** Most systems of this kind do not fail technically. They fail because the person who built them left, and the first time something behaved oddly nobody could explain it, so the team went back to the spreadsheet. The thing actually keeping a system alive is somebody who answers the phone and already understands the question. That is the part the hourly rate is least able to express, and the part Allied will feel the most.

So the rate is set where it is on purpose. Above the freelance floor, because what is being bought is not just the build — it is availability from somebody who is invested in Allied getting this right. Well below the consultancy band, because I do not have a sales team, an account manager or an office to pay for, and because I would rather this get approved in the room than go upstairs to die in a procurement queue.

> **Why that matters:** There is a straightforward self-interest here worth saying out loud, because it should make you trust the number more rather than less. If this works, Allied runs better and I am the person who built the thing that did it. If it does not, I am still here, and everybody knows whose it was. That is a stronger guarantee than any support contract, and it is the one thing a firm you found online genuinely cannot offer you.

Where the hours go

Stage	Work	Hours	Fee
1	Install it; load and check your three lists; set up staff logins; set up and test backups	25	$2,400

Stage	Work	Hours	Fee
2	Sage file production; match your existing order numbering; produce a reconciliation report; establish which Sage you run	17	$1,600
3	Connect and secure the board; two training sessions; written guide; thirty days of free fixes	21	$2,000
	Total	**63**	**$6,000**

What you are not paying for

The 30,061 lines above were written before this job and are not in the price. Neither is the review that found and fixed 59 faults, nor the 150 automatic checks. You are paying for connection, data and training.

Support, priced separately: $350 a month

Alerting if it stops, security updates, backups actually restored and checked, and two hours of changes. Two hours at the hourly rate is $190, so the rest buys availability rather than activity. It is optional.

> **Why that matters:** Most of the value of a support arrangement is that somebody notices a problem before you do. A system that quietly stopped three days ago costs more than the retainer, because nobody discovers it until an order goes missing.

2. What the $6,000 buys back

A price is only meaningful against what it returns, so here is the other side of it. One assumption, stated openly: an order absorbs about **two hours** of someone's day today — reading it, keying it into Sage, deciding the supplier split, raising the purchase orders, assembling the packing slip and certificates, updating the board. Correct that figure if it is wrong and every number here moves with it.

What the system takes instead, measured

A whole order, start to finish, was measured at **25 thousandths of a second**. A full line-by-line check of a supplier bill, dispute letter drafted, at **8 thousandths**. What is left for a person is reading it and saying yes — **about five minutes**. The machine will never be the thing anyone is waiting on.

Orders a week	Hours back a year	Worth at $30 an hour	Pays back the $6,000 in
5	498 h	$14,950	21 weeks
10	997 h	$29,900	10 weeks

Orders a week	Hours back a year	Worth at $30 an hour	Pays back the $6,000 in
20	1,993 h	$59,800	5 weeks
40	3,987 h	$119,600	3 weeks

$30 an hour is office time fully loaded — wage, tax and overhead, not the wage alone. At twenty orders a week the hours returned come to roughly one full-time person's working year.

> **Why that matters:** Judge this on the payback period rather than the yearly figure, because the payback period is what tells you the size of the bet. At ten orders a week the cost is recovered before the quarter closes. The question in the room is therefore not whether $6,000 is a lot of money — it is whether you believe the two hours.

How much further the busy work can be driven down

Connecting it does not remove everything. Roughly **14 minutes** an order remains: forwarding the order email, clicking import in Sage, emailing the purchase orders, and the five-minute review. Reading supplier bills, previously the largest item at eight minutes, has since been built.

Stage	Hand-work left	Busy work removed	What it costs
Today, by hand	120 min	0%	—
Once it is plugged in (this proposal)	14 min	88%	$6,000, this proposal
Plus collecting mail and sending it	7 min	94%	about a week, quoted separately
Plus writing straight into Sage	5 min	96%	included in stage 2b, if your Sage allows it

> **Why that matters:** The reason those last three are cheap is that the expensive part is already paid for. Teaching a system your parts, your suppliers, your customers and your rules is the work. Once that exists, pointing it at a mailbox is plumbing. This is the opposite of how most software spending goes, where the first thing is cheap and every addition costs more than the last.

3. How the two connections actually work

Worth understanding only because it explains why one stage is cheap and the other is not.

Monday — built to be connected to

Monday publishes a documented way for outside software to read and write its boards, and a way for the board to notify outside software when something changes. Every notification carries a tamper-proof seal, so the system can tell a genuine board update from a forged one and reject anything that does not match.

Connecting it means generating a key and registering an address. Setup, not building.

> **Why that matters:** That is why stage 3 is two to three days rather than two weeks.

Sage — not built to be connected to

Sage 50 offers no published way for outside software to write into it. It does not appear on Sage's own developer site at all. There are only two routes:

Route	How	Needs	Status
Import files	The system writes the files; a person imports them in Sage	Any version of Sage 50	Included
Direct writing	The system writes into the Sage data on the same machine	The desktop version of Sage, on a computer left on	Optional extra

Why direct writing is not promised

The newer online Sage keeps your books on Sage's servers, so there is nothing local to write into. If that is what Allied runs, direct writing is impossible at any price. A one-minute check on your machine settles it before any money changes hands.

> **Why that matters:** This also tells you something useful about anyone else who pitches you. Nobody has seamless Sage 50 integration, because Sage does not offer one. They are either doing import files, as we are, or they have not checked.

4. What you pay to keep it running

Software you already buy

The system adds no licence of its own. These are here for completeness, not as new costs.

	Plan	Price	
Monday	Basic	$9 a seat a month	Enough
Monday	Standard	$12 a seat a month	Recommended
Monday	Pro	$19 a seat a month	Not needed
Sage 50	Pro	$132.17 a month	You have this
Sage 50	Premium	$182.50 a month	You have this

	Plan	Price	
Sage 50	Quantum	$271.17 a month	You have this

The $840 a year you can save, with the arithmetic

Monday limits how often outside software may talk to it: **1,000 times a day** on the cheaper plans, 10,000 on Pro. The system uses about **two per order** — one to read the change, one to write the result back. That covers roughly **500 orders a day**. Allied is nowhere near that, so Standard is enough. At ten seats, Standard instead of Pro saves **$840 a year**.

The system itself, measured on a real machine

	Measured	What that means
Memory used	102 MB	Less than one browser tab
Storage per order	about 25 KB	Ten years of trading fits on a memory stick
Time to start	6 seconds	Restarting is not disruptive
Internet needed	None	Runs on your own network
Cost per order	$0	No usage charges of any kind

Yearly running cost	Year one	After
A computer (an existing office PC works)	$0	$0
Or a dedicated small one, if preferred	$500–900	$0
Electricity	$15–30	$15–30
Backup storage	$0–70	$0–70
The AI assistant, running here	$0	$0
Total, without optional support	**$15–100**	**$15–100**

5. If the assistant moved to a paid online service

The system answers questions using an AI model that runs on your own machine, free. The alternative is a service that charges per question. This prices it properly.

First, almost nothing here uses AI

Reading orders, matching parts, splitting between suppliers, checking supplier invoices, auditing mill certificates, producing packing slips and customs papers — all of it runs on fixed rules with **no AI involved at all**. Document reading is pattern matching, not AI. The only thing that uses a model is the optional assistant you can ask questions to. **If nobody ever used it, every check in this system would still work and the AI cost would be zero.**

Why that matters: This matters more than the price. The parts people worry about with AI — that it guesses, that it is confidently wrong, that it changes its mind — cannot touch your money or your certificates here, because those paths never ask a model anything.

What one question costs

These services charge by the amount of text sent and returned. We captured the actual text the system sends across eight kinds of question — past pricing, metallurgy, welding procedure, supplier terms, customer records, certificates, part lookup, order status — rather than guessing. A typical question sends about three-quarters of a page of background along with it.

Quality of model	One question	10 a day	40 a day	150 a day
Basic	$0.0003	$0.07	$0.28	$1.06
Good	$0.0064	$1.41	$5.63	$21.12
Better	$0.0128	$2.82	$11.26	$42.24
Best available	$0.0320	$7.04	$28.16	$105.60

Monthly figures, 22 working days. Based on rates published 24 October 2026.

A year, at a realistic 40 questions a day

Quality of model	Per year	Years of paying before it equals a $900 computer
Basic	$3.38	266 years
Good	$67.58	13 years
Better	$135.17	7 years
Best available	$337.92	3 years

On money alone, the online service wins

At your likely usage the assistant would cost **between three and seventy dollars a year**. Running a capable model on your own machine needs far more memory than the rest of the system does — the difference between an ordinary office PC and a $900 to $1,500 one. The online service removes that purchase entirely.

The reason to say no anyway

Using an outside service means the assistant's questions leave your building — and those questions carry the background the system attached: customer names, what you quoted, your margins, your supplier terms, your order history.

> **Why that matters:** You have been told this system keeps your information on your premises, and that is currently true. It is also the one promise that is very hard to take back. Once your pricing and supplier terms sit in another company's system, the terms under which they stay there are theirs to change, not yours.

Three honest options:

- **Keep it here.** Choose this if “does any Allied pricing leave the building?” has to stay no. Cost: a better computer, once.
- **Move it online.** Choose this if that is acceptable and you want better answers. Cost: a few dollars a month.
- **Split it.** General technical questions go online; anything touching customer pricing stays here. Most work to build, and only worth it if the first two are both unacceptable.

Where these figures could be wrong

The amount of text per question was measured, but the length of the answers is projected from what the system produces today; a chattier model would cost somewhat more. The figures also assume 40 questions a day, which is a guess about your team, not a measurement. If this decision turns on the money, a week of real use would settle it exactly — but at these amounts, it probably does not.

Allied Universal Operations Bridge — What It Costs, and Why — 24 October 2026. Work figures counted from the system itself. Running figures measured on a real machine. Question sizes captured from live use. Supplier prices read from monday.com, sage.com and the AI vendor's own published page on 24 October 2026, and subject to change by them.